

RAJARATA UNIVERSITY OF SRI LANKA

Department of ICT | Faculty of Applied Sciences

ICT 1209 – WEB TECHNOLOGIES

MINI PROJECT FINAL REPORT

Project Title:	EduNest – Interactive Student Resource Hub
Selected Theme:	Student Resource Hub
Course Unit & Batch:	ICT 1209 Web Technologies 2024 Batch (First Year)
GitHub Repository:	https://github.com/GeethakaBandara/EduNest-Student-Resource-Hub
Submission Date:	September 2026
Group Members:	1. B.G.H. Bandara (Index No: 2701) 2. U.G.S.S. Udagedara (Index No: 2793)

1. Project Overview

1.1 Purpose & Vision

EduNest is an interactive, database-driven web application designed to streamline academic resource sharing among ICT undergraduates at Rajarata University of Sri Lanka. The platform enables students to browse, search, and access lecture materials, tutorial guides, and past examination papers categorized systematically. It serves as a centralized hub to improve peer-to-peer collaboration, resource accessibility, and structured knowledge distribution within the university environment.

1.2 Target Audience

- **Primary Users:** ICT undergraduates seeking organized course materials, tutorials, and past papers.
- **Secondary Users:** Academic staff and peer mentors who wish to publish and distribute learning content efficiently.

1.3 Key Features

- **1. Secure User Authentication:** User registration with BCrypt password hashing, session management, and secure login/logout workflows.
- **2. Categorized Resource Management:** Dynamic resource browsing with filtering options based on course categories (e.g., Notes, Tutorials, Past Papers).
- **3. Dynamic Client-Side Search & Validation:** JavaScript-driven real-time search filtering, smooth scrolling, and instant client-side form validation.
- **4. Interactive User Dashboard:** Personal user panel for registered students to manage their uploaded contributions and account details.
- **5. Contact & Query System:** A dynamic feedback and contact form that captures user queries directly into the MySQL database.

2. System Design

2.1 Directory & Folder Structure

The project strictly complies with the standardized file and folder structure mandated by the ICT 1209 course guidelines:

```
EduNest/├── css/│   └── style.css           # Custom styling and layout overrides├── js/│   └── main.js             # Client-side JavaScript interactive features├── images/│   └── uploads/           # Stored assets and resource thumbnails├── includes/│   ├── db.php             # Database connection handler (PDO/MySQLi)│   ├── functions.php      # Utility PHP functions & input sanitization│   ├── register.php       # User registration handler│   ├── login.php          # Authentication & session initiation│   ├── logout.php         # Session destruction & redirect│   └── contact.php        # Contact form interface & message handler
```

```
index.php          # Home page with resource showcase & dynamic search|—
dashboard.php      # User control dashboard|— README.md      # Project
documentation and setup instructions|— database.sql          # Exported MySQL
database relational schema
```

2.2 Database Schema & Entity Relational Design

The backend MySQL database (edunest_db) consists of three core relational tables:

1. users Table (User Accounts)

- `id` (INT, PRIMARY KEY, AUTO_INCREMENT) - Unique user identifier
- `username` (VARCHAR(50), UNIQUE, NOT NULL) - Account login username
- `email` (VARCHAR(100), UNIQUE, NOT NULL) - User email address
- `password` (VARCHAR(255), NOT NULL) - Hashed password using BCrypt
- `created\_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP) - Account creation timestamp

2. messages Table (Contact Submissions)

- `id` (INT, PRIMARY KEY, AUTO_INCREMENT) - Message ID
- `name` (VARCHAR(100), NOT NULL) - Sender full name
- `email` (VARCHAR(100), NOT NULL) - Sender contact email
- `message` (TEXT, NOT NULL) - Contact message query content
- `created\_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP) - Submission timestamp

3. resources Table (Theme-Specific Domain Table)

- `id` (INT, PRIMARY KEY, AUTO_INCREMENT) - Resource item ID
- `title` (VARCHAR(150), NOT NULL) - Resource title/heading
- `category` (VARCHAR(50), NOT NULL) - Resource category (Notes, Tutorials, Papers)
- `description` (TEXT) - Brief description of material
- `file\_path` (VARCHAR(255), NOT NULL) - Uploaded document file reference
- `user\_id` (INT, FOREIGN KEY referencing users.id) - Contributor reference
- `created\_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP) - Upload timestamp

3. Implementation & Feature Breakdown

3.1 Frontend Layout & CSS (Bootstrap 5)

- Built using responsive Bootstrap 5 components including fluid grid layouts, navigation bars, cards, modals, and carousel items.
- Ensured full responsiveness adapting seamlessly across mobile, tablet, and desktop viewports.
- Implemented consistent navigation bar and footer elements across all pages for cohesive user navigation.

3.2 Client-Side JavaScript Features

- **Feature 1 - Dynamic Content Updates:** Live client-side searching and filtering of academic resources without full page reloads.
- **Feature 2 - Form Validation:** Real-time feedback on user registration and contact forms for email formatting and required field checks.
- **Feature 3 - Smooth Scrolling & Event Handling:** Interactive hover effects, dynamic modal popups, and smooth anchor scrolling.

3.3 Backend & Database Security Integration

- **1. Prepared Statements:** Prepared Statements: Utilized PDO / MySQLi prepared queries across all database operations to eliminate SQL Injection risks.
- **2. Secure Authentication:** Password Hashing: Stored user passwords securely using PHP's password_hash() with BCRYPT algorithm.
- **3. Session Management:** Session Defense: Executed session_regenerate_id(true) immediately after successful login to prevent session fixation vulnerabilities.

SCREENSHOT PLACEHOLDER: Please insert actual UI screenshots of index.php, dashboard.php, login.php, and contact.php into this section before final submission.

4. Challenges and Solutions

Challenge Encountered	Root Cause	Implemented Solution
SQL Injection Vulnerability	Direct concatenation of user inputs into raw SQL query strings.	Implemented PDO/MySQLi prepared statements with bound parameters across all CRUD operations.
Session Fixation Attack Risk	Re-using pre-login session tokens across authenticated session states.	Applied session_regenerate_id(true) upon successful authentication in login.php.
Responsive Layout Distortions	Table and resource cards overflowing on narrow mobile viewports.	Leveraged Bootstrap 5 responsive grid columns (col-12 col-md-6 col-lg-4) and flexbox wrappers.
Duplicate Form Submissions	Page refreshing re-sending POST data payloads to the server.	Implemented Post/Redirect/Get (PRG) pattern following database INSERT execution.

5. Individual Contribution Breakdown

Geethaka Bandara

- Configured the relational MySQL database schema (database.sql) and PDO database connection handler (includes/db.php).
- Implemented secure backend user authentication (register.php, login.php, logout.php) using BCrypt password hashing and session regeneration.
- Integrated prepared statements across all backend PHP workflows to ensure complete SQL injection protection.
- Managed project Git version control, repository structure, and commit history on GitHub.

[Group Member 2 Name]

- Designed interactive UI/UX frontend components using HTML5, CSS3, and Bootstrap 5 framework.
- Developed client-side Vanilla JavaScript scripts for dynamic resource search filtering, form validation, and scroll animations.
- Implemented contact.php interface and database insertion logic for user feedback messages.
- Drafted project report documentation, user guide, and created architectural wireframe diagrams.

6. Setup & Execution Instructions (XAMPP)

Follow these step-by-step instructions to run the application locally:

- **Step 1:** Launch XAMPP Control Panel: Start both the Apache Web Server and MySQL Database modules.
- **Step 2:** Database Import: Open browser and navigate to <http://localhost/phpmyadmin/>. Create a new database named `edunest_db`. Click the Import tab, select `database.sql` from the project directory, and click Go.
- **Step 3:** Deploy Project Files: Copy the entire project folder into the XAMPP web root directory: `C:\xampp\htdocs\EduNest`.
- **Step 4:** Verify Database Connection: Inspect `includes/db.php` to verify credentials (`$host = 'localhost', $user = 'root', $pass = '', $dbname = 'edunest_db'`).
- **Step 5:** Execute Application: Launch browser and navigate to <http://localhost/EduNest/index.php>.

7. References

[1] PHP Documentation Team, PHP Manual: Password Hashing & Prepared Statements, Available at: <https://www.php.net/docs.php>

[2] Bootstrap Team, Bootstrap v5.3 Framework Documentation, Available at: <https://getbootstrap.com/docs/5.3/>

[3] Mozilla Developer Network (MDN), JavaScript Web Docs & DOM API Reference, Available at: <https://developer.mozilla.org/>

[4] Rajarata University of Sri Lanka, ICT 1209 Web Technologies Mini Project Guide 2024 Batch, Department of ICT.